

LOKESH R M

AI / ML Research Engineer · Embodied AI, Multi Agent Systems, Robot Learning

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SUMMARY

AI and machine learning research engineer working on embodied AI, multi agent systems and robot autonomy. Current work at Sirena Technologies spans simulation to real transfer for legged robots, LLM based agent systems with a policy enforcement layer, and vision and speech for human robot interaction. Independent work on federated learning under differential privacy and homomorphic encryption.

RESEARCH INTERESTS

Embodied AI · Robot Learning · Multi Agent Systems and Agent Governance · Simulation to Real Transfer · Federated and Decentralised Learning · Differential Privacy · Human Robot Interaction

EDUCATION

B.Tech, Computer Science and Engineering - GITAM (Deemed to be University), Bengaluru

Aug 2022 - Aug 2026

CGPA: 7.9 / 10

SELECTED RESEARCH PROJECTS

Agent OS and Policy Gateway, Sirena Technologies

2026

Multi agent platform of 53 services allowing heterogeneous in house and third party agents to interoperate and model organisational functions. Designed the current architecture and built the Policy Gateway, a governance layer enforcing 153 policies with centralised authorisation and monitoring across all agents.

Stack: Python, LLMs, LangGraph, MCP, Open Weight Models, pgvector, Kubernetes

Secure Federated Learning with Blockchain

Sep 2024

Federated training protocol letting organisations train a shared model on decentralised datasets without exchanging raw records. Applied differential privacy and homomorphic encryption to client updates, with a permissioned blockchain making aggregation rounds auditable and recording data provenance.

Stack: Python, TensorFlow, Hyperledger Fabric, Solidity, Homomorphic Encryption, Differential Privacy

Simulation to Real Transfer for Legged Robots, Sirena

2025 - 2026

Built MATLAB and Python locomotion simulations for biped and quadruped platforms covering walk sequences and control algorithms, validating every robot in simulation before hardware build and carrying the work through to physical controllers.

Stack: MATLAB, Python, Gazebo, NVIDIA Isaac Sim, Isaac Lab, MuJoCo, ROS 2

Action Recognition on Drone Video, MURTI Labs

2024 - 2025

Investigated LSTM based architectures for object detection and action recognition on aerial video, reaching 83% accuracy, and built the labelled data and image processing pipeline used to train and evaluate them.

Stack: Python, TensorFlow, OpenCV, NumPy

EXPERIENCE

AI Engineer - Sirena Technologies, Bengaluru, India

Jul 2026 - Present

- Architected Agent OS and own the Policy Gateway governance layer; led 5 in house agents and 3 third party integrations. Platform team of 12, gateway team of 3.
- Led a 5 person team building the Sirena Cloud service layer supporting 80+ concurrent robots on an NVIDIA RTX Pro 6000, covering voice, vision and LLM reasoning.
- Built a streaming voice pipeline at roughly 60 ms response latency, and fine tuned YOLO and SSD detectors for office and industrial environments.

Robotics Engineer, Graduate Trainee - Sirena Technologies, Bengaluru and Solan, India

May 2025 - Jun 2026

- Locomotion simulation and control for biped and quadruped robots, the LS6 robot controller codebase, and ROS 2 motion control for robotic arms and wheeled robots.
- Led a 3 person team building the cloud system for an ESP32 P4 based home robotics product. Completed the graduate rotation at Solan and converted to a full time role.

Machine Learning Research Intern - MURTI Labs, GITAM Bengaluru, India

May 2024 - Apr 2025

- Applied machine learning research on autonomous delivery and emergency response drones: object detection, action recognition, path mapping and the supporting data pipeline.

TECHNICAL SKILLS

Machine Learning: PyTorch, TensorFlow, Scikit-learn, LSTM, LLMs, RAG, Multi Agent Systems, LangGraph, MCP, Federated Learning, Differential Privacy, Homomorphic Encryption

Perception and Robotics: YOLO, SSD, OpenCV, ONNX, ASR, TTS, ROS 2, Nav2, SLAM, Gazebo, NVIDIA Isaac Sim, Isaac Lab, MuJoCo, MATLAB, Sim2Real, Jetson Orin Nano

Programming and Systems: Python, C++, C, MATLAB, SQL, Linux, Docker, Kubernetes, FastAPI, Google Cloud, PostgreSQL, Git

AWARDS

- INSPIRE Award MANAK** (Government of India, state level) · **Young Scientist Award** (Government of Karnataka, district level) · **Rajya Puraskar** (Governor's Award) · Finalist, Internal Smart India Hackathon, GITAM
- Secretary, IEEE Computer Society GITAM** (Dec 2024 - Aug 2025) · Student Placement Coordinator, GITAM Career Guidance Cell